



Jyothi Engineering College(Autonomous)

DEPARTMENT OF MECHATRONICS ENGINEERING

Vth Semester - B.Tech

First Internal Examination, Aug 2026

13-Aug-2026

PEMRT525 - IOT AND APPLICATIONS

Total Mark: 30

CODE: MRU5X130826F

Total Time: 1 Hrs : 15 Mins

CO1	Apply IoT fundamentals to model and explain IoT architectural views	Applying(P)
CO2	Apply knowledge of smart objects, sensors, actuators, and functional blocks in IoT systems	Applying(P)
		CO BL MARK

PART A

1. Explain the Publish-Subscribe communication model. Identify the three roles and the protocol that implements it. State one key advantage over Request-Response. CO1 2 (3)
2. Identify which IoT functional block is responsible for authentication, encryption, and data integrity. Explain why this block is described as a 'cross-cutting concern' across all other blocks. CO1 2 (3)
3. Define a smart object. List its four components. Describe how combining these components creates the 'smart' in a smart object. CO2 2 (3)
4. Identify and describe the criteria you would consider when planning a WSN deployment for a smart agriculture field monitoring system. CO2 2 (3)

PART B

5. Apply your knowledge of three IoT enabling technologies — Wireless Sensor Networks (WSN), Cloud Computing, and Big Data Analytics — to demonstrate how each addresses a specific IoT challenge. Illustrate how they form an integrated IoT pipeline with a suitable example. CO1 3 (9)

OR

6. Apply the IoT deployment model of Bahga & Madisetti to classify any three deployment levels. Use your understanding to determine the most suitable level for a multi-building campus energy monitoring system and justify your selection. CO1 3 (9)
7. Apply the WSN protocol stack framework to describe IEEE 802.15.4, ZigBee, and 6LoWPAN at each layer. Use five communication criteria to select and justify the most suitable wireless technology for a smart hospital patient monitoring system. CO2 3 (9)

OR

8. Illustrate WSN architecture with a labelled diagram. Apply the FFD and RFD classification to describe node capabilities and their differences. Use knowledge of the six design constraints of WSN nodes to explain how data aggregation and self-organisation address these constraints. CO2 2 (9)